

Advanced Deep Learning Architectures and Adaptive Frameworks for Medical Image Restoration

A Thesis

Submitted in partial fulfilment for the Degree of
Doctor of Philosophy under the
Faculty of Artificial Intelligence

by

Siju K.S



Department of Artificial Intelligence
Faculty of Artificial Intelligence
AMRITA VISHWA VIDYAPEETHAM
COIMBATORE CAMPUS (INDIA)

May 2026

Department of Artificial Intelligence
AMRITA VISHWA VIDYAPEETHAM
COIMBATORE CAMPUS (INDIA)



Bonafide Certificate

This is to certify that the thesis entitled “**Advanced Deep Learning Architectures and Adaptive Frameworks for Medical Image Restoration**” submitted by **Siju K.S**, Roll. No: CB.AI.R4CEN24003, for the award of the Degree of **Doctor of Philosophy** under the **Faculty of Artificial Intelligence** is a bonafide record of the work carried out by him/her under my guidance and supervision at **Department of Artificial Intelligence**, Amrita Vishwa Vidyapeetham, Coimbatore Campus (India).

Dr. Vipin V

Thesis Advisor,
Associate Professor,
Department of Artificial Intelligence,
Amrita Vishwa Vidyapeetham,
Coimbatore Campus (India)

Department of Artificial Intelligence
Amrita School of Engineering
AMRITA VISHWA VIDYAPEETHAM
COIMBATORE CAMPUS (INDIA)

Declaration

I, **Siju K.S**, Roll. No: CB.AI.R4CEN24003, hereby declare that this thesis entitled “**Advanced Deep Learning Architectures and Adaptive Frameworks for Medical Image Restoration**” is the record of the original work done by me under the guidance of **Dr. Vipin V**, Associate Professor, Department of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus (India). To the best of my knowledge, this work has not formed the basis for the award of any degree/diploma/associateship/fellowship or a similar award to any candidate in any University.

Place: Coimbatore

Date:

Signature of the Scholar

Countersigned

Dr. Vipin V

Thesis Advisor,
Department of Artificial Intelligence,
Amrita School of Engineering,
Amrita Vishwa Vidyapeetham, India.

*To my parents, family, and teachers,
whose guidance and blessings made this journey possible.*

ACKNOWLEDGEMENTS

I express my sincere gratitude to my thesis advisor, **Dr. Vipin V**, and co-advisor, **Dr. Mithun Kumar Kar**, for their invaluable guidance, constant encouragement, and unwavering support throughout this research journey.

I extend my heartfelt thanks to the members of my Doctoral Committee for their constructive feedback and scholarly advice. I am grateful to the **Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore Campus (India)** for providing the research facilities and infrastructure necessary to carry out this work.

I am indebted to my fellow research scholars for stimulating discussions and moral support. I owe a deep debt of gratitude to my family for their patience, encouragement, and unwavering belief in me throughout this endeavour.

This work was supported by [**Funding agency / grant details if any**].

Siju K.S
CB.AI.R4CEN24003
Artificial Intelligence
Amrita Vishwa Vidyapeetham
May 2026

ABSTRACT

Medical images acquired using X-ray and computed tomography (CT) are often degraded by acquisition noise and patient motion, reducing diagnostic accuracy. Deep learning methods have shown promise for image restoration, but existing approaches have limitations. Standard denoisers often smooth out fine anatomical details. Most activation functions apply the same operation everywhere, without considering local context.

This thesis addresses these limitations through a three-phase investigation. Phase I presents **X-GAN**, a physics-informed generative adversarial network for chest X-ray denoising, achieving a PSNR of 36.48 dB and an Edge Preservation Index of 0.721 with only 2.17 million parameters. Phase II introduces **SAGA**, a spatially adaptive gated activation function improving PSNR by 6.3 dB over ReLU on average across four architectures. Phase III will develop **Viveka**, a test-time refinement framework, and **ADL₁**, a Sobolev-regularised loss function with an attention-guided U-Net that guarantees gradient fidelity. **Keywords:**

Medical image restoration, deep learning, chest X-ray denoising, image deblurring, generative adversarial network, activation function, test-time refinement, Sobolev loss, edge preservation

CONTENTS

Bonafide Certificate

Declaration

Acknowledgements **i**

Abstract **ii**

List of Figures **v**

List of Tables **vi**

List of Abbreviations **vii**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Problem Statement 1

1.3 Research Objectives 1

1.4 Scope and Limitations 1

1.5 Organisation of the Thesis 1

2 Literature Review **2**

2.1 Overview 2

2.2 Topic Area 1 2

2.3 Topic Area 2 2

2.4 Topic Area 3 2

2.5 Research Gaps 2

2.6 Summary 2

3 Methodology **3**

3.1 Overview of Proposed Framework 3

3.2 Dataset Description 3

3.3 Proposed Architecture 3

3.4 Loss Function 3

3.5	Training Details	3
3.6	Evaluation Metrics	3
4	Results and Discussion	4
4.1	Experimental Setup	4
4.2	Quantitative Results	4
4.3	Ablation Study	4
4.4	Qualitative Results	4
4.5	Discussion	4
5	Conclusion and Future Work	5
5.1	Summary of Contributions	5
5.2	Conclusion	5
5.3	Limitations	5
5.4	Future Work	5
	List of Publications	6
	References	7
A	Supplementary Material	8
A.1	Dataset Statistics	8
A.2	Additional Results	8

LIST OF FIGURES

LIST OF TABLES

3.1	Training hyperparameters	3
4.1	Comparison with state-of-the-art methods	4

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADL ₁	Augmented Differential L ₁ Loss
AF	Activation Function
AG-UNet	Attention-Guided U-Net
BN	Batch Normalisation
CBAM	Convolutional Block Attention Module
CBDNet	Convolutional Blind Denoising Network
CLAHE	Contrast-Limited Adaptive Histogram Equalisation
CT	Computed Tomography
DCT	Discrete Cosine Transform
DL	Deep Learning
DnCNN	Denoising Convolutional Neural Network
DXA	Dual-Energy X-ray Absorptiometry
ECS	Edge Concentration Score
EDSR	Enhanced Deep Super-Resolution
ELU	Exponential Linear Unit
EPI	Edge Preservation Index
FID	Fréchet Inception Distance
GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GLCM	Grey-Level Co-occurrence Matrix
LRP	Layer-wise Relevance Propagation
MAE	Mean Absolute Error
MOS	Mean Opinion Score
MSE	Mean Squared Error
NIQE	Naturalness Image Quality Evaluator
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
ResNet	Residual Network
RIDNet	Real Image Denoising Network

Abbreviation	Full Form
SAGA	Spatially Adaptive Gated Activation
SMAPE	Symmetric Mean Absolute Percentage Error
SSIM	Structural Similarity Index Measure
U-Net	Encoder-Decoder Network with Skip Connections
VGGNet	Very Deep Convolutional Network
X-GAN	Physics-Informed Generative Network for X-ray
X-ReCNN	X-ray Residual Channel Neural Network

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Medical imaging is essential for modern diagnosis, treatment planning, and disease monitoring. X-ray and computed tomography (CT) are the most widely used modalities because they are fast, accessible, and cost-effective. In low-income countries with limited access to MRI or ultrasound, X-ray remains the primary diagnostic tool for conditions ranging from fractures to pneumonia.

However, image quality is often degraded during acquisition, reducing diagnostic accuracy. This can lead to missed findings, false positives, or unnecessary repeat scans. Poor image quality also forces clinicians to request additional views or higher radiation doses, increasing patient risk.

1.2 Problem Statement

1.3 Research Objectives

1.4 Scope and Limitations

1.5 Organisation of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 presents a comprehensive literature review. Chapter 3 describes the proposed methodology. Chapter 4 presents experimental results and discussion. Chapter 5 concludes the thesis and outlines future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter surveys existing work relevant to the research objectives stated in Chapter 1.

2.2 Topic Area 1

2.3 Topic Area 2

2.4 Topic Area 3

2.5 Research Gaps

Based on the survey above, the following gaps are identified.

1. **Gap 1:** ...
2. **Gap 2:** ...
3. **Gap 3:** ...

2.6 Summary

CHAPTER 3

METHODOLOGY

3.1 Overview of Proposed Framework

3.2 Dataset Description

3.3 Proposed Architecture

3.4 Loss Function

3.5 Training Details

Table 3.1: Training hyperparameters

Parameter	Value
Optimiser	Adam
Learning rate	1×10^{-4}
Batch size	16
Epochs	100

3.6 Evaluation Metrics

The following metrics are used to evaluate performance.

PSNR Peak Signal-to-Noise Ratio measures pixel-level fidelity.

SSIM Structural Similarity Index captures perceptual similarity.

EPI Edge Preservation Index quantifies retention of high-frequency structural information.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Experimental Setup

All experiments were conducted on a system with . . .

4.2 Quantitative Results

Table 4.1 compares the proposed method with state-of-the-art baselines.

Table 4.1: Comparison with state-of-the-art methods

Method	PSNR (dB)	SSIM	EPI	Params (M)
Baseline A	–	–	–	–
Baseline B	–	–	–	–
Proposed	–	–	–	–

4.3 Ablation Study

4.4 Qualitative Results

4.5 Discussion

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

This thesis has made the following contributions:

1. **Contribution 1:** ...
2. **Contribution 2:** ...
3. **Contribution 3:** ...

5.2 Conclusion

5.3 Limitations

5.4 Future Work

LIST OF PUBLICATIONS

Journal Articles (Published)

1. Siju K.S, [Co-author(s)]. Title of paper. *Journal Name*, Vol(Issue):pages, Year. DOI: <https://doi.org/...>

Journal Articles (Communicated / Under Review)

1. Siju K.S, [Co-author(s)]. Title of paper. Submitted to *Journal Name*, Year.

Conference Papers

1. Siju K.S, [Co-author(s)]. Title of paper. In *Proceedings of Conference Name*, pages, Year.

REFERENCES

CHAPTER A

SUPPLEMENTARY MATERIAL

A.1 Dataset Statistics

A.2 Additional Results